


Most of the bacterial cells contain cell wall which is,partially composed of a macromolecule called peptidoglycans, a mixture of amino sugars and short peptides. Human cells do not need peptidoglycans and do not make them. Certain antibiotics act by targeting such cell walls and peptidoglycans. Penicillin, one of the first antibiotics widely used to prevent the final cross-linking step, or transpeptidation as a result of which the fragile cell wall would burst, killing the bacterium.

## Plant cell wall

Algae have cell wall, made of cellulose, galactans, mannans and minerals like calcium carbonate, while in other plants it consists of cellulose, hemicellulose, pectins and proteins. The cell wall of a young plant cell, the primary wall is capable of growth, which gradually diminishes as the cell matures and the secondary wall is formed on the inner (towards membrane) side of the cell. The middle lamella is a layer mainly of calcium pectate which holds or glues the different neighbouring cells together. The cell wall and middle lamellae may be traversed by plasmodesmata which connect the cytoplasm of neighbouring cells.

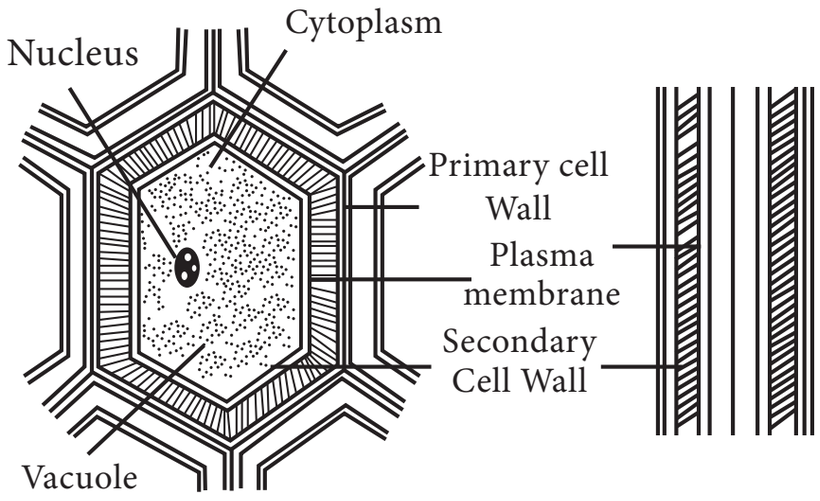



*Figure 1.5.A Schematic representation of a typical plant cell wall*

## The main functions of the cell wall are:

- Cell wall provides structural and mechanical support.
- Cell wall determines and maintains the shape of the plant cell and governs plant architecture



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